DBMS: Normalisation

Normalisation is the process of organising a relational database to reduce redundancy and avoid insert, update and delete anomalies. A table is in First Normal Form (1NF) if every column holds atomic values. It is in Second Normal Form (2NF) if it is in 1NF and every non-key attribute depends on the whole primary key, not on part of it.

It is in Third Normal Form (3NF) if it is in 2NF and no non-key attribute depends on another non-key attribute (no transitive dependency). Boyce-Codd Normal Form (BCNF) is a stricter form of 3NF in which every determinant is a candidate key.